

CL677 : Modelling Stochastic and Turbulent Transport

(Spring 2025-26)

Assignment 3

(total points: 47+5 = 52)

Important Note: Students are expected to actively collaborate with your group members and solve this assignment. Each group has to submit one PDF file with your detailed answers to this assignment. There will be an online viva session in which each member will be assessed individually; every member is expected to have an indepth understanding of the submitted work and to be able to explain/justify any step in the calculation or code. The individual marks awarded during the viva will be used to scale the marks awarded to the group-submission. Copying across groups is prohibited; if detected, both groups involved in copying (the lender and the copier) will be penalized with zero marks. So do not share codes or answers across groups.

Question A: Consider the following stochastic differential equation (SDE), which corresponds to the Fokker-Planck equation (FPE) of Q. B in Assignment 2.

$$dx = -ax \, dt + dW.$$

Solve this SDE using the Euler-Maruyama method for an ensemble of trajectories (start with 100, increase to 1000, and more if needed), beginning with a uniform initial distribution between $[-1, +1]$ (chosen to match the initial condition of the corresponding FPE. Evolve the equations until stationarity is attained (you can identify this by monitoring $\langle x^2 \rangle$), and then evolve it further to gather statistics in the stationary state. Store values of x from all trajectories at intervals of time of order $1/a$ (to ensure that the data collected is statistically informative).

1. Solve the SDE for $a = 0.2, 0.5, 1$. Plot the stationary p.d.f of x . For this you will have to use a grid of x values to obtain intervals or bins of x . Then you can count the frequency of occurrences for each bin and normalize with the total number of events. This will give you the probability for each bin. Dividing by the bin size will then yield values of the pdf for the mid-point of each interval. This numerical approximation of the p.d.f. is basically a normalized histogram and is a simple consequence of its definition. The approximation can be refined by choosing smaller bin sizes, but this will in turn require more statistics to get a converged result. In other words, for a chosen bin size, you will have to construct the p.d.f with an increasing amount of data (using a longer time in the stationary state or more trajectories) until convergence. (20)
2. Overlap the p.d.f.s obtained from the SDE with the result of solving the FPE. The match should be very good, if the domain for the FPE is large enough to mimic an unbounded domain, and if the statistics from the SDE are good enough. (5)

Question B: The following SDE with a space-dependent Brownian force is a simple and important case of multiplicative stochastic forcing and is called a geometric random walk.

$$dx = x \, dW \tag{B1}$$

$$dx = x \circ dW \tag{B2}$$

Here, (B1) is the Ito interpretation while (B2) is the Stratonovich interpretation. These two equations will yield different results because the noise is multiplicative.

1. Obtain the Ito equivalent of (B2). Derive analytically the evolution of the $\langle x \rangle$ and $\langle x^2 \rangle$ for both equations using Ito calculus (4).
2. Simulate an ensemble of trajectories for both (B1) and (B2). Let all trajectories start at unity. Compare the numerical and analytical results for the evolution of $\langle x \rangle$ and $\langle x^2 \rangle$ for both equations. Simulate the trajectories for a long enough time for a clear comparison. (10)
3. Compare the p.d.f at different times for both equations and discuss the results. Can you anticipate the differences by comparing (B1) and the Ito version of (B2)? (6)
4. Which of the two kinds of dynamics would you expect to see for a physical system, in which the SDEs are obtained by replacing fast, unknown degrees of freedom with Brownian noise? Explain. (2)

Writing and plotting: 5 additional marks are allotted for evaluating the comprehensiveness of the work and the quality of the report. Well-reasoned discussions and explanations of results, and clear and elegant plots will be rewarded. (5)